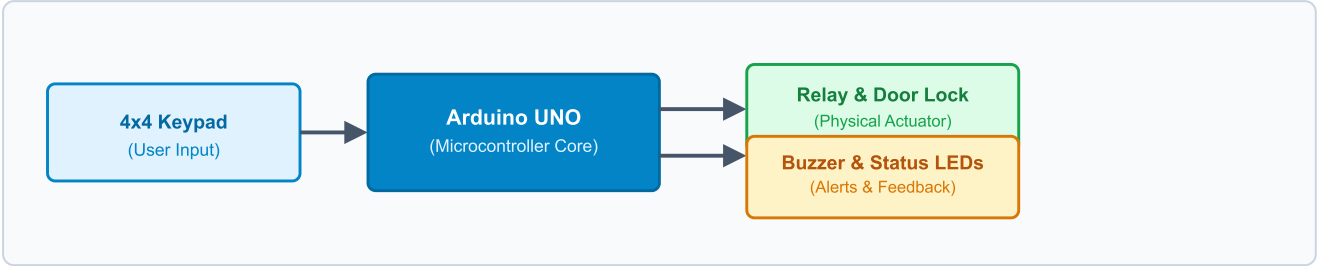
PROJECT ABSTRACT WRITE-UP

ARDUINO-BASED SMART KEYPAD ACCESS CONTROL SYSTEM

METHODOLOGY

- **Input Capture:** The system utilizes a 4x4 matrix membrane keypad connected to digital pins of the microcontroller to process user passcode entries.
- **Processing & Verification:** An Arduino microcontroller reads keypresses, compares the inputted passcode against pre-configured security credentials stored in EEPROM memory, and evaluates authorization instantly.
- **Actuation & Control:** Upon successful passcode validation, the controller triggers a 5V relay module driving a solenoid door lock or servo motor mechanism to unlatch the door.
- **Alert & Feedback System:** Visual status is indicated via LEDs, while repeated invalid attempts trigger a local piezoceramic buzzer alarm to prevent unauthorized brute-force entry.

PROPOSED BLOCK DIAGRAM



PROPOSED OUTCOME

The anticipated result of this project is a fully functional, low-cost, and reliable physical hardware prototype for automated door access control. Key outcomes include:

- Elimination of physical keys, mitigating security risks associated with unauthorized key duplication or loss.
- Enhanced physical access security for restricted environments such as server rooms, academic laboratories, and offices.
- A highly responsive standalone hardware architecture requiring minimal maintenance and low operational power consumption.
- Flexible design enabling future IoT extension modules (such as Wi-Fi/Bluetooth) for real-time access logging and remote monitoring.